YUTING HE

Environmental Engineering | Statistics | GIS & NatCat Risk | Applied AI

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M.Sc. candidate in Environmental Engineering (expected February 2027), combining statistics with geospatial risk analytics, applied machine learning and AI-assisted development. Experience spans reinsurance, hydrology research, engineering teaching and transparent decision-support prototypes.

EDUCATION

M.Sc. Environmental Engineering - candidate
Oct 2023 - expected Feb 2027

Technical University of Munich (TUM)

Hydrology and water resources, environmental risk, uncertainty analysis and machine learning in forecasting.

B.Sc. Statistics; minor in Computer Science
Oct 2020 - Feb 2025

Ludwig Maximilian University of Munich (LMU)

Statistical modelling, machine learning, databases, algorithms and programming. Final grade: 1.89.

B.Sc. Environmental Engineering
Oct 2018 - Jun 2023

Technical University of Munich (TUM)

Water engineering, hydrology, hydraulics, wastewater systems, sustainable design and transportation systems.

PROFESSIONAL EXPERIENCE

Intern - Hydrological Risk
Nov 2025 - Aug 2026

Munich Re - Geo Risk, Corporate Underwriting

- Prepared and quality-checked geospatial flood hazard and exposure datasets for NatCat analysis.
- Built ArcPy/Python geoprocessing workflows and a MATLAB interface for hydrological risk analysis.
- Used Microsoft Copilot and LLM tools for recurring workflows and technical documentation.

Student Research Assistant - Machine Learning in Hydrology
Apr 2025 - Jul 2025

TUM - Chair of Hydrology and River Basin Management

- Prepared Python/Google Colab teaching notebooks covering preprocessing, feature engineering, training and evaluation for hydrological forecasting.

Teaching Assistant - Hydromechanics
Oct 2022 - Feb 2024

TUM - Professorship of Hydromechanics

- Led exercise sessions, explained quantitative fluid-mechanics solutions and prepared structured feedback on recurring mistakes.

Landscape Planning Assistant (full-time)
Apr 2017 - Nov 2017

Earthasia Design Group, China

- Assisted residential, agricultural and commercial landscape planning, using AutoCAD, Adobe Photoshop/Illustrator and MS Office.

YUTING HE

Selected projects and technical methods

SELECTED PROJECTS

HeatLens Germany - Multi-Hazard Decision Translation

2026

- Built a Germany-wide GIS research application with 614 hydrological sub-basins, 400 district views and 16 state summaries, joined by exact area-overlap weights.
- Integrated DWD/ICON ensembles, RADOLAN observations, UFZ soil-moisture percentiles and DWD crop-specific soil-water context.
- Implemented role-specific checks, freshness gates, shareable views and English/German explanations. Scores are uncalibrated screening indices, not official warnings.

JavaScript | Python | GIS | NetCDF/HDF5 | Automated data and browser checks

Live demo | Source

GridScout - Adaptive Power Grid Inspection

Oct 2026

- Developed a Bayesian decision-support prototype for adaptive drone inspection after extreme weather, using a 64-segment Bavaria grid map.
- Created a mission replay and auditable evidence dashboard with synthetic telemetry and visual-inspection evidence. Demonstration probabilities are uncalibrated.

Python | Streamlit | NetworkX | Bayesian inference | JavaScript

Public showcase | Source

Spatiotemporal Drought Prediction

Mar 2025 - Jul 2025

- Developed drought classification models from climate, soil, land-use and terrain features; compared LSTM, Transformer and GNN approaches with temporal and spatial validation.
- Extended the work with a CAMELS-DE Neckar catchment graph for one-month-ahead drought context.

Python | PyTorch | scikit-learn | pandas | NumPy | GIS | Graph learning

Additional environmental and NGO prototypes

Jun 2026

- Forest degradation in Ethiopia: open geospatial visualization and a restoration-prioritization prototype for NGO scenarios.
- Impact Atlas: AI-assisted curation and ranking of news, reports, funding calls and field signals by NGO mission, region and topic.

TECHNICAL TOOLKIT

GIS and risk	ArcGIS Pro, QGIS, ArcPy; raster/vector processing, exposure data checks, flood and drought analysis.
Programming	Python, R, pandas, NumPy, scikit-learn, PyTorch basics, SQL, Java, basic C++, Git; JavaScript web prototypes.
Hydrology tools	HEC-HMS, WaSiM, HYDRO-AS 2D, MODFLOW and MATLAB.
Reporting	Excel, PowerPoint, LaTeX, matplotlib, seaborn, ggplot2, Power BI and Tableau.
AI-assisted work	Microsoft Copilot, GitHub Copilot, ChatGPT, Claude and Codex for prototyping, debugging and documentation.

LANGUAGES

Chinese - native | English - C1 | German - C1